

Severity Matrix

Objective

The AI SOC Copilot assesses the overall impact of detected cybersecurity threats by assigning a **Risk Score (0–100)** and a **Severity Level (Low, Medium, High, or Critical)**. These values are generated by the **Google Gemini Large Language Model (LLM)** after analyzing the parsed security logs, extracted Indicators of Compromise (IOCs), attack context, and observed malicious behavior. This enables security analysts to prioritize incidents based on their potential impact and urgency.

Severity Assessment

The AI model evaluates multiple aspects of an incident before assigning a severity level and risk score. The assessment considers the attack type, extracted IOCs, MITRE ATT&CK mapping, confidence level, and overall threat context to determine the potential impact of the incident.

For documentation and analyst interpretation, the following risk score ranges can be used to understand the AI-generated results.

Severity	Risk Score Range	Description	Recommended Analyst Action
Low	0–25	Minor suspicious activity with minimal impact.	Monitor the event and continue routine observation.
Medium	26–50	Potential security incident requiring further investigation.	Validate the alert and perform additional analysis.
High	51–75	Confirmed malicious activity affecting one or more systems.	Initiate containment procedures and investigate immediately.
Critical	76–100	Active compromise or high-impact attack requiring urgent response.	Escalate the incident, isolate affected systems, and begin full incident response.

Note: The AI SOC Copilot does **not** use hardcoded thresholds to calculate severity. Instead, the Gemini LLM generates both the **Risk Score** and **Severity Level** dynamically based on the security logs and extracted IOCs. The score ranges above are provided only as guidance for analysts when interpreting the AI-generated results.

Factors Considered During Severity Assessment

The AI evaluates multiple indicators before determining the severity of an incident, including:

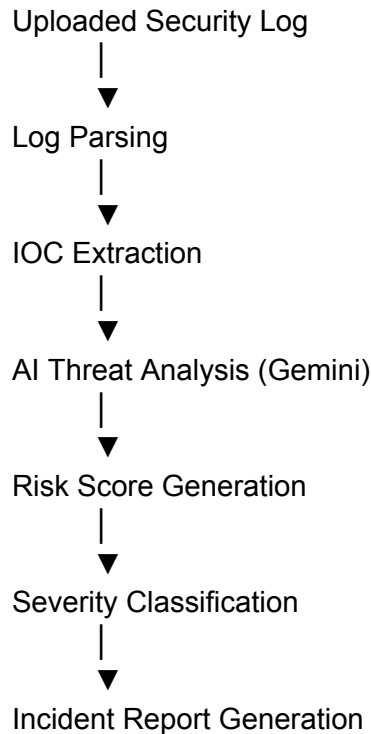
- Attack type identified from the security logs
 - Extracted Indicators of Compromise (IOCs)
 - MITRE ATT&CK tactic and technique
 - Potential business impact
 - Evidence of malicious execution or privilege escalation
 - Presence of suspicious network activity or command-and-control communication
 - Overall confidence of the AI analysis
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Example Severity Assessment

The following example represents a typical output generated by the AI SOC Copilot.

Field	Value
Attack Type	Advanced Persistent Threat (APT)
Severity	Critical
Risk Score	95/100
Confidence	97%
MITRE ATT&CK	T1059.001 – PowerShell
Tactic	Execution

Severity Assessment Workflow



Importance of Severity Assessment

The severity assessment module enables SOC analysts to:

- Prioritize incidents based on their potential impact.
 - Focus on high-risk threats requiring immediate attention.
 - Reduce alert fatigue by distinguishing critical incidents from low-priority events.
 - Support faster incident response and decision-making.
 - Provide standardized threat prioritization for security operations.
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Conclusion

The AI SOC Copilot uses Google Gemini to analyze security logs, extracted IOCs, and attack context to generate a Risk Score and Severity Level for each incident. By combining AI-driven threat analysis with MITRE ATT&CK mapping and IOC extraction, the system helps SOC analysts quickly identify high-priority threats and respond effectively to cybersecurity incidents.